Computational Fluid Dynamics for Engineering Applications

Assignment: Finite volume calculations : Due 25th February 2026

Two very large parallel plates, with an incompressible fluid in the space between them, are separated by a distance h. One of the plates is suddenly accelerated from rest and moves at a constant velocity U_o while maintaining the same distance from the other plate, which remains stationary. The following governing equation describes the development of the fluid velocity profile in the direction perpendicular to the plates with time:

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2}$$

where u is the fluid velocity in the direction parallel to the plates, ν is the kinematic viscosity of the fluid, t is time, and y is the distance in the direction perpendicular to the plates. The boundary and initials conditions are:

$$\begin{array}{lll} t \le 0: & u = 0 & \text{for all } y \\ t > 0: & u = U_o & \text{for } y = 0 \\ & u = 0 & \text{for } y = h \end{array}$$

The analytical solution of the governing equation, which satisfies these boundary and initial conditions, is:

$$u(y,t) = \frac{U_o}{h}(h-y) - \frac{2U_o}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\left(\frac{n\pi}{h}y\right)e^{-\left(\frac{n^2 t}{\lambda}\right)} \qquad \text{where} \quad \lambda = \frac{h^2}{\pi^2 \nu}$$

Assume that the distance between the plates is 10 mm, the velocity of the moving plate is 1 m/s, and the fluid is water with dynamic viscosity 10^{-3} kg/ms and density 1000 kg/m^3.

1) (i) Divide the domain into 5 finite volumes and discretise the governing equation at each node using the explicit Euler discretisation scheme.

(i) Using a time step of 0.2 s, solve your set of equations to determine the development of the fluid velocity profile between the plates with time.

(ii) Compare your numerical solution with the analytical solution given above at 1, 5, 10 s, and comment on any differences, providing percentage errors to back up your statements.

(ii) Show what condition must be satisfied in order to achieve a stable solution. Demonstrate graphically (using your numerical solver) what happens when this condition is not met.

2) (i) Once again, divide the flow domain into 5 finite volumes, and discretise the governing equation at each node using the Crank-Nicolson scheme.

(ii) Using a time step of 0.2 s, solve your set of equations to determine the development of the fluid velocity profile between the plates with time. Compare your numerical solution with the analytical solution at 1, 5, 10 s, and comment on any differences, providing percentage errors to back up your statements.

(iii) What is the maximum grid spacing and time step you recommend to achieve a solution that is independent of these quantities? Explore at least three different time steps and three different grid spacings. Plot graphs of grid spacing and time-step versus percentage errors to support your statements.

Calculations may be done by writing a program in C/C++/Matlab or Python.

Please submit a short report with **a maximum of 5 pages** including graphs and title page (but excluding appendices). Your explicit and implicit discretisations should be placed in an appendix at the end of the report, together with tables that summarise the algebraic equations for the boundary and internal nodes of each discretisation scheme. Another appendix should contain your program. These must be documented (formulas must be shown in the case of spreadsheets, and programs must be commented) sufficiently well that the calculations can be understood from the submitted copy alone.

Due: 25/02/2026